The surface of Earth is constantly moving, very slowly, and has been doing so for more than three billion years. The lithosphere (Earth's crust and upper mantle) is divided into seven large sections and many smaller ones, called tectonic plates. Where plates meet, the type of movement determines the nature of the boundary. Where plates push against each other, new mountains are created. If plates pull apart, new crust forms on the ocean floor.

The first inkling that the continents may not have always been in their current positions came in the late 16th century. European explorers and colonizers sailing to the Americas saw from their newly created maps that the coastlines on each side of the Atlantic Ocean mirrored each other. Later, geologists found strong structural and geological similarities between the Caledonian-era mountains of Northern Europe and the Appalachian Mountains of North America.

Lookalike fossils

There are various examples of fossil finds straddling different continents that can only be explained by continental movement—since the animals or plants concerned would have been unable to cross the ocean divide. These include *Cynognathus crateronotus*, a mammal-like reptile that lived over 200 million years ago in southern Africa and eastern South America. *Glossopteris*, a genus of woody trees, grew in South America, South Africa, Australia, India, and Antarctica, but nowhere else, around 300 million years ago.

To German geophysicist Alfred Wegener, such fossil patterns indicated that these continents had once been joined together. In 1915, he published his theory that all the continents were once a single land mass, "Pangaea," which has since broken up and drifted apart.